

4.1.3 Cells in animals and plants

Understanding the structure of cells, the transport of substances into and out of cells, cell division by mitosis and meiosis and cell differentiation lays the foundations for the study of systems in the human body in [Systems in the human body](#) (page 36), of plant biology in [Plants and photosynthesis](#) (page 42) and of inheritance in [Inheritance](#) (page 79).

There are two required practicals: an activity observing cells under a light microscope and an investigation of the effect of different concentrations of salt or sugar solutions on plant tissues.

Microscopes are used to study cells and so practical work can include the microscopic examination of plant and animal cells.

4.1.3.1 Electron microscopy

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how electron microscopy has increased our understanding of sub-cellular structures.	An electron microscope has a much higher resolving power than a light microscope. This means that it can be used to study cells in much finer detail. An electron microscope can magnify up to a million times ($\times 1\,000\,000$) or more, which is much more than a light microscope which has a useful magnification of only about a thousand times ($\times 1\,000$). $\text{magnification} = \frac{\text{size of image}}{\text{size of real object}}$	MS 2a, 2h Demonstrate understanding of number, size and scale and the quantitative relationship between units. WS 4.5 Interconvert units. MS 1a,1b, 1c, 2h Carry out calculations involving magnification, real size and image size including numbers written in standard form. WS 3.3 Carry out and represent mathematical and statistical analysis. WS 4.6 Use an appropriate number of significant figures. WS4.4 Use prefixes centi, milli, micro and nano. MS 1d, 2h Make order of magnitude calculations. MS 1d Use estimations and explain when they should be used.

4.1.3.2 Cell structures

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the main sub-cellular structures of eukaryotic cells (plants and animals) and prokaryotic cells are related to their functions, including the nucleus/genetic material, plasmids, mitochondria, chloroplasts and cell membranes.	Plant and animal cells (eukaryotic cells) have a cell membrane, cytoplasm and a nucleus containing the genetic material. Bacterial cells (prokaryotic cells) are much smaller in comparison. They have cytoplasm and a cell membrane surrounded by a cell wall. The genetic material is not enclosed in a nucleus. It is a single DNA loop and may have one or more small rings of DNA called plasmids. Most animal cells have the following parts: • a nucleus • cytoplasm • a cell membrane • mitochondria • ribosomes. Most human cells are like most other animal cells. In addition to the parts found in animal cells, plant cells often have: • chloroplasts • a permanent vacuole filled with cell sap. Plant and algal cells also have a cell wall made of cellulose, which strengthens the cell.	

Required practical activity 3: use a light microscope to observe, draw and label a selection of plant and animal cells. A magnification scale must be included.

AT skills covered by this practical activity: biology AT 1 and 7.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 164).

4.1.3.3 Transport into and out of cells

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how substances are transported into and out of cells through diffusion, osmosis and active transport.	Some substances move across cell membranes via diffusion. Diffusion is a spreading out and mixing process. Particles move from a region where they are in higher concentration to a region where their concentration is lower. Factors that affect the rate of diffusion across a membrane are: • the difference in concentration • the temperature • the surface area of the membrane. Water may move across cell membranes by osmosis. Cell membranes are partially permeable: they allow small molecules such as water through but not larger molecules. During osmosis water diffuses from where it is more concentrated (because the solute concentration is lower), through a partially permeable membrane to where water is less concentrated (because the solute concentration is higher). Some substances move across cell membranes via active transport. Active transport involves the movement of a dissolved substance from a region where it is less concentrated to a region where it is more concentrated. This requires energy from respiration. Active transport allows mineral ions to be absorbed into plant root hairs from very dilute solutions in the soil. It also allows sugar molecules to be absorbed from lower concentrations in the gut into the blood with a higher sugar concentration.	MS 4a, 4b, 4c, 4d Plot, draw and interpret appropriate graphs. WS 3.4 Represent the distribution of results and make estimations of uncertainty. MS 1c Calculate percentage gain and loss of mass. WS 3.3 Carry out and represent mathematical and statistical analysis.

Required practical activity 4: investigate the effect of a range of concentrations of salt or sugar solutions on the mass of plant tissue.

AT skills covered by this practical activity: biology AT 1, 3 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 165).

4.1.3.4 Mitosis and the cell cycle

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the process of mitosis in growth, including the cell cycle.	The nucleus of body cells contains chromosomes. In body cells the chromosomes are normally found in pairs. There are 46 chromosomes in human body cells. DNA is in the chromosomes and each chromosome carries a large number of genes. Cells divide so that organisms can grow during the development of multicellular organisms, and repair damaged tissues. Dividing cells go through a series of stages called the cell cycle. During the cell cycle the genetic material doubles and then divides to give two new cells that are genetically identical to each other and to the original cell. Knowledge of the stages of the cell cycle and mitosis is not required. Before a cell can divide it must grow, and make copies of all the organelles such as mitochondria and ribosomes. It must also replicate the chromosomes in the nucleus. Then it can divide by mitosis. During mitosis, the two complete sets of chromosomes are pulled to opposite sides of the cell. Two new nuclei form. Then the cell splits into two.	

4.1.3.5 Meiosis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the role of meiotic cell division in halving the chromosome number to form gametes.	Cells in reproductive organs divide by meiosis to form gametes (egg and sperm cells). Knowledge of the stages of meiosis is not required. When a cell divides to form gametes: • copies of the genetic information are made • the cell divides twice to form four gametes, each with a single set of chromosomes • all gametes are genetically different from each other. Gametes join at fertilisation to make a new cell with the normal number of chromosomes. The new cell divides by mitosis to grow.	

4.1.3.6 Cell differentiation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the function of stem cells in embryonic and adult animals. Explain the importance of cell differentiation.	At first the cells in an embryo can grow and divide to form any type of cell. They are stem cells. As an embryo develops most of the cells differentiate and become specialised. Specialised cells carry out a particular function. Differentiation is essential to produce a variety of cells with different functions in multicellular organisms (animals and plants). Cells that have become specialised cannot later change into different kinds of cells. However, there are some stem cells in most adult tissues that are ready to start dividing to replace old cells or to repair damage in the tissues where they are found.	This section links with Stem cells (page 66).